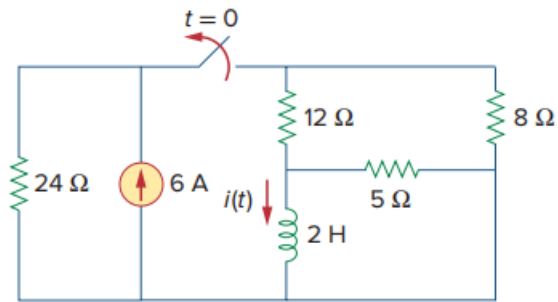


Problem

For the circuit in Fig. 7.18, find $i(t)$ for $t > 0$.

Fig. 7.18:



Step-by-step solution

Step 1 of 9

Consider the following circuit diagram:

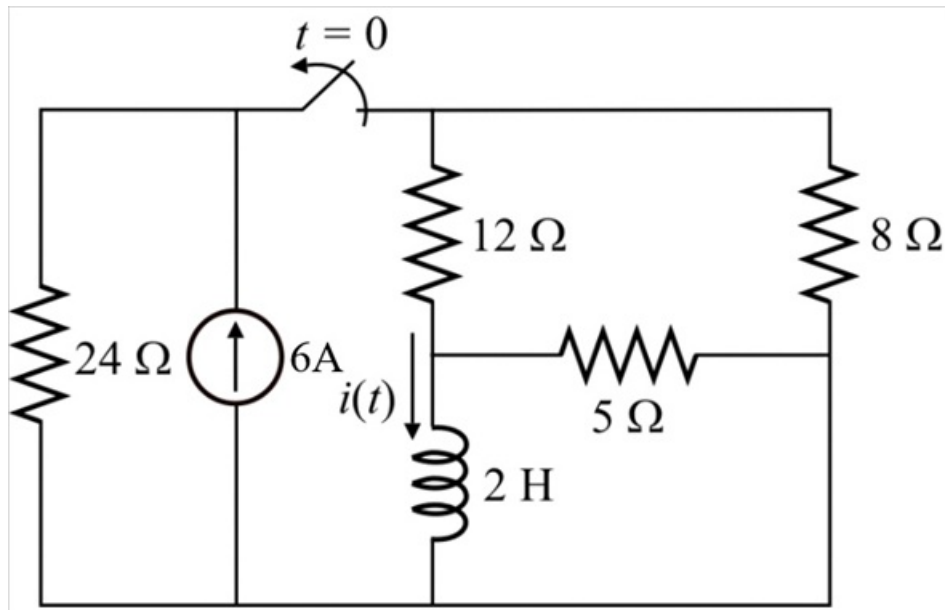


Figure 1

Step 2 of 9

At $t = 0^-$, switch in the circuit is closed and inductor acts as short circuit. Draw the modified circuit diagram:

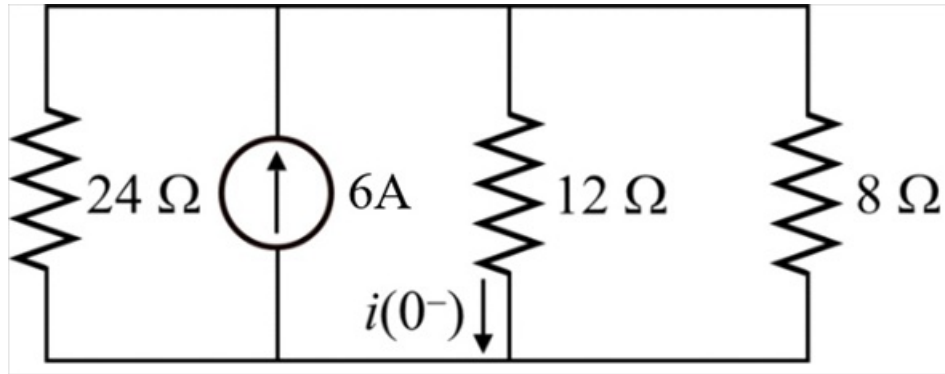


Figure 2

Step 3 of 9

Resistors $24\ \Omega$ and $8\ \Omega$ in the circuit shown in Figure 2 are connected in parallel. Calculate the equivalent resistance.

$$\begin{aligned} R &= (24) \parallel (8) \\ &= \frac{(24)(8)}{24+8} \\ &= \frac{192}{32} \\ &= 6\ \Omega \end{aligned}$$

Step 4 of 9

Draw the modified circuit after converting parallel resistors into equivalent.

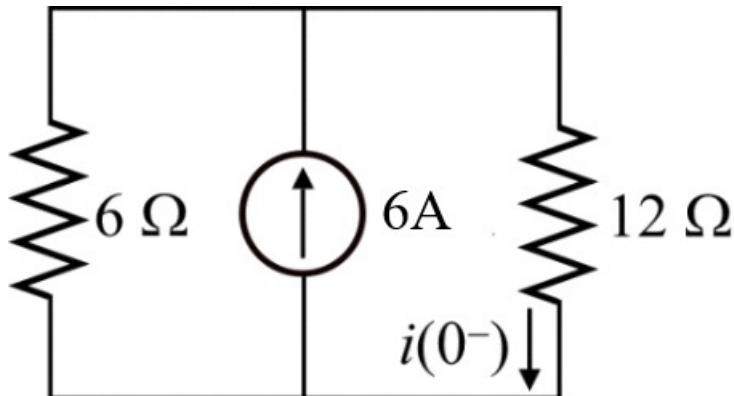


Figure 3

Step 5 of 9

Calculate the initial current through the inductor.

$$\begin{aligned} i(0^-) &= \left(\frac{6}{6+12} \right) (6) \\ &= \left(\frac{6}{18} \right) (6) \\ &= \frac{6}{3} \\ &= 2 \text{ A} \end{aligned}$$

The current through inductor cannot change suddenly.

$$i(0^-) = i(0^+) = i(0) = 2 \text{ A}$$

Step 6 of 9

For $t > 0$, the switch in the circuit is open and inductor acts as a short circuit. Draw the modified circuit diagram:

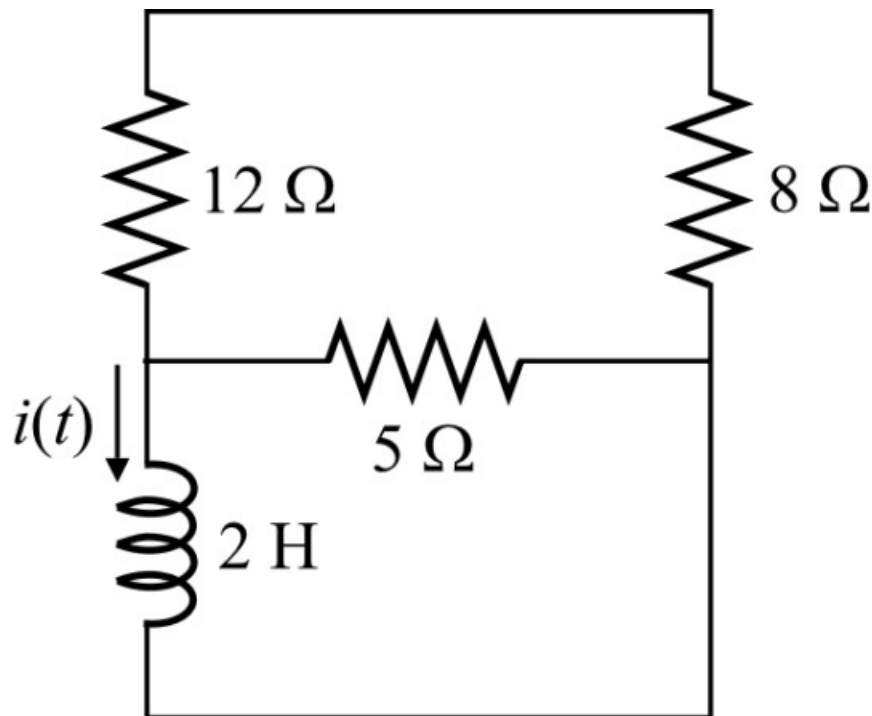


Figure 4

Step 7 of 9

The circuit shown in Figure 4 is a source free RL circuit.

Write the general expression for source free RL circuit for $t > 0$.

$$i(t) = i(0) e^{-\frac{t}{\tau}} \dots\dots (1)$$

Looking from the terminals of the inductor shown in Figure 4, resistors 12Ω and 8Ω are connected in series and this series combination is in parallel with 5Ω resistor. Calculate the equivalent resistance.

$$\begin{aligned} R_{eq} &= (12+8) \parallel (5) \\ &= \frac{(20)(5)}{20+5} \\ &= \frac{100}{25} \\ &= 4 \Omega \end{aligned}$$

Step 8 of 9

Calculate the time constant of RL circuit.

$$\begin{aligned}\tau &= \frac{L}{R_{\text{eq}}} \\ &= \frac{2}{4} \\ &= \frac{1}{2} \text{ s}\end{aligned}$$

Step 9 of 9

Substitute $\frac{1}{2}$ s for τ and 2 A for $i(0)$ in the equation (1).

$$\begin{aligned}i(t) &= (2)e^{-\frac{t}{\left(\frac{1}{2}\right)}} \\ &= 2e^{-2t} \text{ A}, t > 0\end{aligned}$$

Hence, the current through inductor, $i(t)$ is $2e^{-2t} \text{ A}, t > 0$.